People Analytics Project

TransitIQ

Employee Attrition Analysis

Identifying Key Drivers & Recommendations to Reduce Attrition from 16% to 10%

PT. TransitIQ | Deleonard Simanjorang | 2026

Our Company !

TransitIQ

Who Are We?

TransitIQ is a logistics and transportation company specializing in the delivery of oversized goods and heavy equipment, including trucks (tronton) and trailers serving clients across Indonesia.

Our Mission

- Deliver every shipment on time and safely to every customer, without exception.
- Provide best-in-class logistics services and become the #1 transportation and logistics company in Indonesia.

Executive Summary

TransitIQ is currently **losing Rp 4.74 billion** annually due to a **16% attrition rate**. By hitting our 1**0% target**, we will save **Rp 1.8 billion** every year in recruitment costs alone.

Employees who work overtime have an attrition rate of **31%**, nearly **3x higher** than non-overtime employees, indicating **overtime is a primary attrition driver** across all departments.

Early-tenure employees (Year 0–1) face the highest resignation risk at **36%**, indicating **potential mismatches in recruitment**, **onboarding**, or **workload expectations** from day one.

This analysis leverages **MySQL** and **Power BI** to evaluate 1**,470 employee** records, pinpointing how **overtime**, **job roles**, and **tenure** drive workforce trends.

The Sales Department has the highest attrition rate at **21%**, with Sales Representatives reaching **40%**, signaling systemic issues in **workload management** and target-setting.

Three data-driven recommendations are proposed to reduce attrition from **16%** to **10%** within **6 months**, with progress monitored via a real-time Power BI **dashboard.**

We are losing Rp 4.7B/year !

Why solving attrition is a financial priority

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TransitIQ loses an estimated **Rp 4.740.000.000 annually** due to **high employee attrition**, equivalent to **hiring 237 employees every year**.

Rp 20.000.000 Cost per hire (conservative estimate = ⅓ annual salary)	Rp 4.740.000.000 Current annual Recruitment cost at 16 % attrition (237 employees)	Rp 2.940.000.000 Projected annual Cost at 10% Attrition target (147 employees)

Projected Annual Savings: Rp 1.800.000.000 By reducing attrition from 16% to 10% within 6 months

Cost per hire estimated at 1/3 of annual salary (~Rp 60.000.000) based on conservative industry benchmark. Actual costs including productivity loss may be higher.

Analysis Objectives

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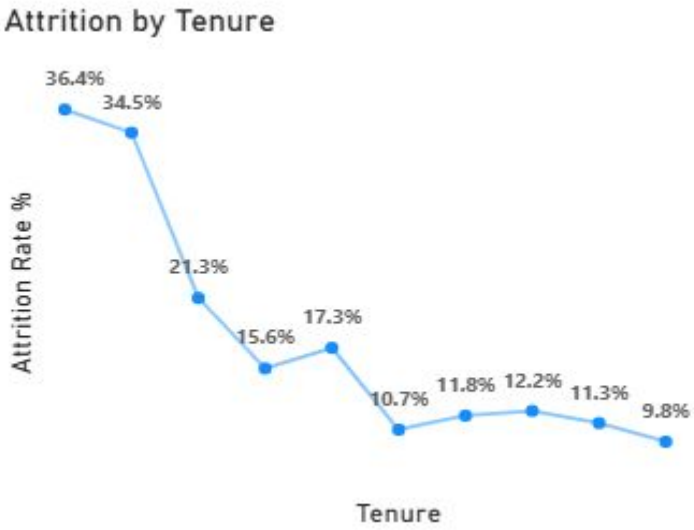
What's driving this? - Project Background



Attrition is most critical among new employees, with Year 0–1 recording rates as high as **36%**, indicating systemic onboarding and engagement gaps.

Benchmark Attrition Rate		
< 10% Healthy Attrition Rate Benchmark	15% Transportation & Logistics	**16%** TransitIQ Current Rate

Reference Benchmark

TransitIQ is currently experiencing an employee attrition rate of **16%**, exceeding the healthy industry benchmark and signaling an urgent need for data-driven intervention.

The **10%** target is based on the industry benchmark for healthy attrition. Achieving this would mean TransitIQ retains **88 additional employees annually**, directly **reducing recruitment and training costs**.

Project Goals

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Problem Statement

TransitIQ is experiencing an employee attrition rate of **16%**, which is **6%** above the healthy industry benchmark of 10%.

Problem Statement

To identify the root causes of employee attrition and provide data-driven recommendations to reduce the attrition rate from **16%** to **10%** within **6 months**, measured through a real-time monitoring dashboard.

In Scope

This analysis utilized the IBM HR Analytics dataset from Kaggle, comprising **1,470** employee records and **35** variables.

Attrition was analyzed across all departments, with key variables including **Overtime**, **Job Role**, **Department**, and **Years at Company** (**Tenure**).

Out Scope

Several variables were excluded from this analysis, including **Age**, **Monthly Income**, and **Performance Rating**, as the focus was placed on operational and structural factors driving attrition.

Tools Used:
MySQL (data exploration, cleaning & analysis), **Power BI** (dashboard development & monitoring)

Methodology

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1 Business Problem Definition

Define business problem & SMART objective

Tools : **MySQL**

2 Data Preparation & Cleaning

Check missing values, duplicates & inconsistent values

Tools : **Python**

3 Data Analysis

Identify attrition patterns & key drivers

Tools : **MySQL**

4 Dashboard Development

Build KPI Monitoring dashboard in Power BI

Tools : **Power BI**

5 Insight & Recommendation

Translate findings into actionable recommendations

Output : **Business Insights**

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Deep Dive Analysis

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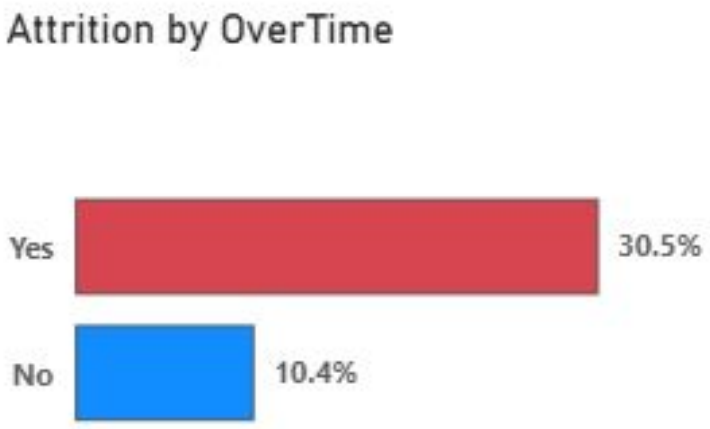
What does data tell us? — Overtime



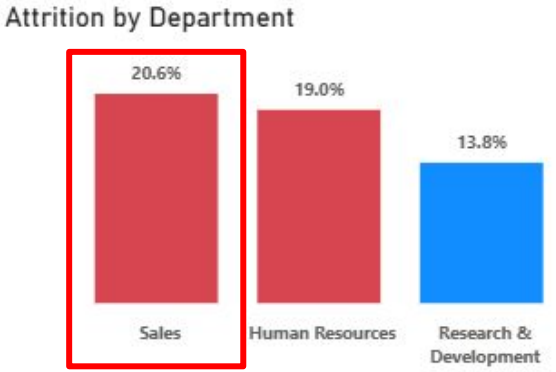
Key Finding :

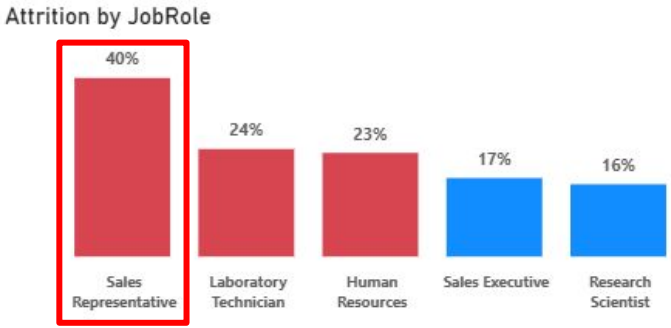
Employees who work overtime have an attrition rate of **31%**, nearly **3x higher** than non-overtime employees at **10%.**

Insight :

Excessive overtime signals poor task allocation and unrealistic targets, not insufficient effort from employees.

What does data tell us? — Department & Job Role





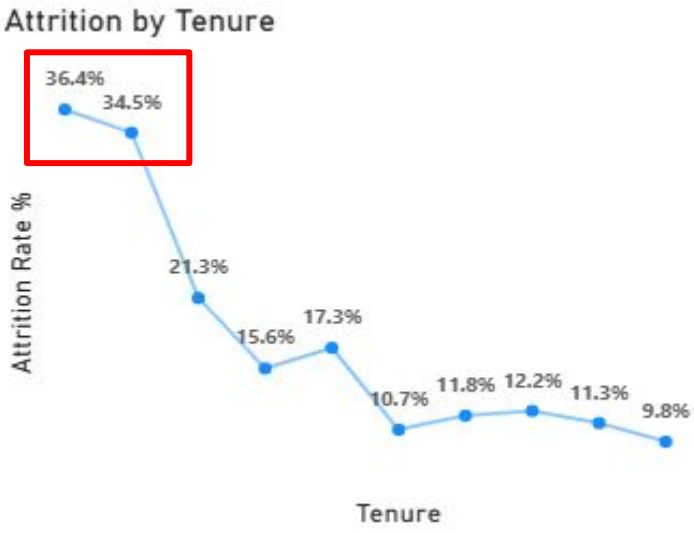
Key Finding :

Sales Department has the highest attrition rate at **21%**, with Sales Representatives reaching **40%**, the **most critical role** in the dataset

Insight :

High attrition in Sales suggests systemic issues with **target-setting** and **workload** management, particularly for frontline sales roles.

What does data tell us? — Employee Tenure



Key Finding :

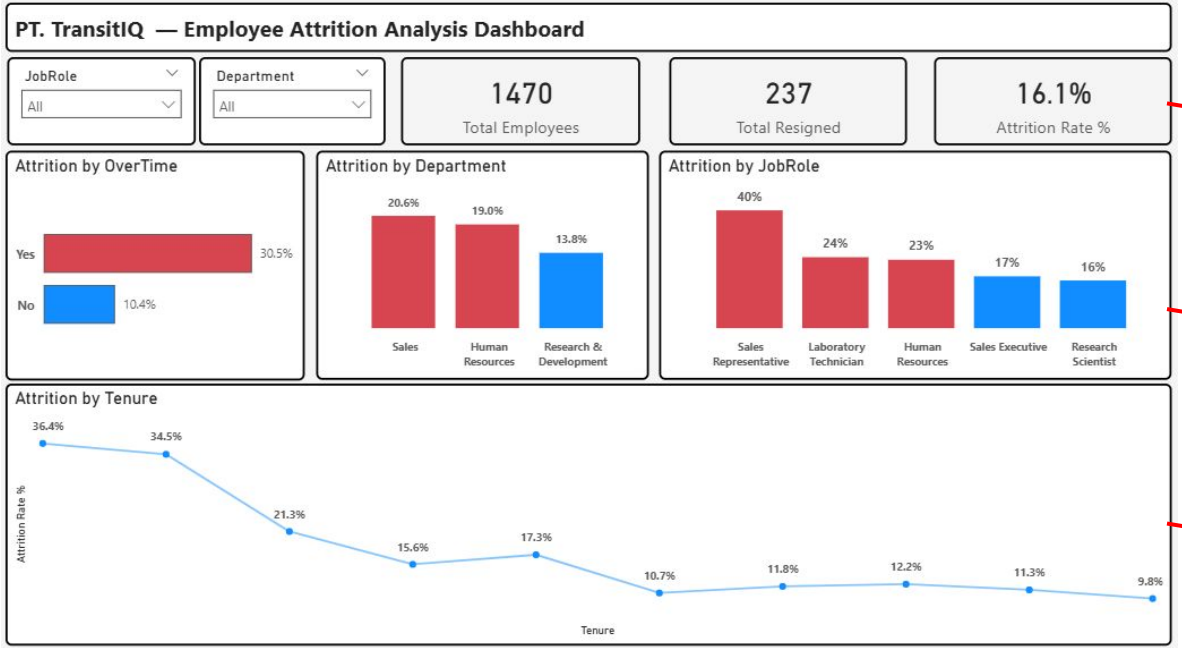
Attrition is highest among employees in their first 2 years, **36%** at Year 0 and **35%** at Year 1, nearly **3x higher** than employees with 5+ years of tenure.

Insight :

High early-tenure attrition signals gaps in onboarding, unclear job expectations, and lack of structured support for new employees.

How do we track progress?

Real-time Attrition Monitoring Dashboard



KPI Cards
Key metrics: Total Employees, Total Resigned & Overall Attrition Rate at a glance

Bar Charts
Attrition breakdown by Overtime, Department & Job Role, identifying highest risk areas

Line Chart
Tenure analysis revealing critical attrition risk in Year 0–1 employees

Dashboard Goal :
Monitor attrition progress toward the 10% target across all key risk areas

Insight & Recommendations

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Overtime

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Key Finding :

Employees who work overtime have an attrition rate of **31%**, nearly **3x higher** than non-overtime employees at 10%. **289 active overtime employees remain at high risk**.

Why This Happens :

Overtime is often a symptom of:

- Unclear task allocation & role overlap
- Non-essential tasks consuming productive hours
- Workforce imbalance, some roles overstaffed, others understaffed
- Culture of "more hours = more productive" that accelerates burnout

Risk if Unaddressed :

289 overtime employees are still active and at high risk of leaving. Attrition will continue to rise, increasing hiring costs and repeating operational inefficiencies.

Recommendation

1. Workload Clarity **Conduct 1-on-1 sessions** per role to map actual workload vs job description. Provide clear task instructions to eliminate role ambiguity and overlap.
   Tool: RACI Framework per role

2. **Workforce Mapping** Analyze headcount distribution per department and job role to identify overstaffed or understaffed areas. Reallocate resources proactively before overtime becomes unavoidable.
   Tool: Headcount optimization model

Sales Department

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Finding :

The Sales Department has the highest attrition rate at **21%**, with Sales Representatives reaching **40%**, nearly **2x higher** than the company average of 16%.

Risk if Unaddressed :

Continued high attrition in Sales will **directly impact revenue generation** and **increase recruitment costs** for **the most critical customer-facing roles**.

Recommendation

1. Workload Clarity Conduct 1-on-1 sessions per role to map actual workload vs job description. Provide clear task instructions to eliminate role ambiguity and overlap.
   Tool: RACI Framework per role

2. Workforce Mapping Analyze headcount distribution per department and job role to identify overstaffed or understaffed areas. Reallocate resources proactively before overtime becomes unavoidable.
   Tool: Headcount optimization model

Expected Impact :

Reduce overtime attrition from 31% → below 15% within 6 months

Overtime

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Key Finding :

Employees in Year 0–1 face the highest attrition risk at 36% — nearly 3x higher than employees with 5+ years of tenure at 10%.

Root Cause Hypothesis :

Data shows high early attrition, but the exact root cause requires further investigation :

Hypothesis 1
Mismatch between job description and actual responsibilities, employees feel misled about the role during recruitment.

Hypothesis 2
Excessive workload from day one, new employees in learning curve are expected to perform at full capacity immediately.

Risk if Unaddressed :

289 overtime employees are still active and at high risk of leaving. Attrition will continue to rise, increasing hiring costs and repeating operational inefficiencies.

Recommendation

1. Validate Root Cause First Implement structured exit interviews for all Year 0–1 resignations to identify the true driver.
   Tool: Exit interview template with standardized questions

2. Realistic Job Preview During recruitment, provide honest day-to-day job expectations not just job description to reduce post-hire mismatch.
   Tool: Pre-hire job shadowing or realistic job preview deck

Early Tenure

Finding :

Employees in their first 2 years face the highest attrition risk, **36% at Year 0** and **35% at Year 1**, nearly **3x higher** than employees, with 5+ years of tenure.

Risk if Unaddressed :

Without **structured onboarding**, new employees will **continue to leave** before becoming **fully productive**, creating a **perpetual cycle of hiring and retraining**.

Recommendation

1. Develop **a structured 30-60-90 day onboarding program** with clear milestones and expectations.
2. **Implement a mentor-mentee system** pairing new employees with senior staff for guidance and support.
3. Conduct **monthly engagement session** during the first year to **identify disengagement early.**

Expected Impact :

Reducing early-tenure attrition from **36%** targeting below **20%** in Year 0–1

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Conclusion & Next Steps

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Conclusion & Next Steps

TransitIQ is losing Rp 4.740.000.000 annually due to high attrition. Three targeted interventions are projected to reduce attrition from 16% to 10% within 6 months, saving Rp 1.800.000.000 in recruitment costs alone.

Month 1 & 2

Implement

Workload audit & RACI framework

Exit interview program

30-60-90 onboarding rollout

Month 3 & 4
Monitor

Track attrition rate via dashboard

Identify Early Warning signals

Validate hypothesis

Month 5 & 6
Evaluate

Measure attrition reduction vs 10% target

Adjust strategies if needed

Report ROI to stakeholder

What's the ROI? :

Reduce attrition 16% to 10% | Save Rp 1.800.000.000 | Monitored via Dashboard

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THANK YOU

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Appendix

Appendix A — References & Data Source

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Dataset

IBM HR Analytics Employee Attrition Dataset

Source : Kaggle

Link : kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Records : 1,470 employees

Variables: 35 columns

Industry Benchmark References

1. Insignia Resources (2025) "Average Turnover Rate by Industry" → Healthy attrition benchmark: < 10%

2. firstPRO (2025) "2025 Employee Turnover: What's Average?" → Transportation & Logistics avg: ~13–15%

3. U.S. Bureau of Labor Statistics (2024) → Overall U.S. turnover rate: 3.4–3.6%

Appendix B — Key SQL Queries

MySQL queries used for analysis

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```
-- Overall Attrition Rate

SELECT
Attrition,
COUNT(*) AS total_employee,
ROUND(COUNT(*) * 100/ SUM(COUNT(*)) OVER(), 2)
AS attrition_rate_pct
FROM hr_attrition
GROUP BY Attrition;
```

```
-- Attrition by Tenure

SELECT
YearsAtCompany,
COUNT(*) AS total_employee,
SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS
resigned,
ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0
END) * 100 / COUNT(*), 2) AS attrition_rate_pct FROM
hr_attrition
GROUP BY YearsAtCompany;
```

```
-- Attrition by Overtime

SELECT
OverTime,
COUNT(*) AS total_employee,
SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS resigned,
ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) AS attrition_rate_pct
FROM hr_attrition
GROUP BY OverTime;
```

Appendix C — About the Analyst

Deleonard Simanjorang
Data Analyst | People Analytics Enthusiast

Background in HR & Business Operations, transitioning into data analytics with focus on workforce analytics, people analytics, and operational efficiency.

Technical Skills SQL (MySQL, BigQuery) · Python · Power BI Tableau · Excel · Data Visualization

Certifications
· Business Intelligence Analyst — Digital Skola (2026)
· Full Stack Data Analytics — RevoU (2025)
· Microsoft Excel Mastery — Digital Skola (2026)

Links
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